

Abstract

Heavy metal pollution is affecting the marine environment of the Puget Sound. This study examined the physiological and immunological stress responses of exposure to lead and copper on Purple Shore Crabs (*Hemigrapsus nudus*). Twelve crabs were randomly assigned to control, lead, or copper treatments, and starved for three weeks at 14°C and 35 ppt. Righting time, total hemocyte count, and osmolality were used as indicators of stress, immune response, and osmoregulatory function. Righting time was analyzed by one-way ANOVAs within each treatment group across weeks, while total hemocyte count and mean osmolality were analyzed by one-way ANOVAs across treatment groups. No significant differences were found for righting time or hemocyte count (Control: $p=0.0649$, Lead: $p=0.326$, Copper: $p=0.729$; $p=0.244$). Osmolality also did not differ significantly among all treatment groups ($p=0.0633$), but a separate one-way ANOVA showed a significant difference between the lead and control groups ($p=0.0065$). These results suggest that lead exposure may affect osmoregulation in Purple Shore Crabs, while copper and lead may contribute to behavioral and immune stress responses. While not statistically significant, the results are likely due to a small sample size. From these results, future studies should include more replicates or additional metal concentrations to better evaluate the physiological effects of heavy metal exposure on shore crabs.

Introduction

Although home to a plethora of marine life, the Puget Sound is mainly known for the variety of shipping companies it houses. While beneficial to the economy, shipping ports are detrimental to the marine environment as they emit large amounts of pollution into the water. These pollutants include pesticides, sewage, and most importantly, heavy metals (Jiang et al., 2025). The top metals found in the water are aluminum, arsenic, cadmium, chromium, copper, iron, lead, manganese, mercury, nickel, selenium, and zinc (Swam-Jaramillo et al., 2025). These metals are commonly put into the water through chemical waste, fertilizers/pesticides, and anti-fouling paint (Sherwood Lollar & Holland, 2007; Turner, 2010).

In Washington state, more specifically, the Puget Sound, there are numerous types of taxa that reside in these waters. Among this marine life is the Purple Shore Crab (*Hemigrapsus nudus*). The Purple Shore Crab is a decapod crustacean in the family *Varunidae*. This species is known to be a strong osmoregulator and is able to withstand harsh, diverse environments (Lee & Ko, 2008).

When exposed to heavy metals, marine organisms may experience increased stress responses such as slowed growth rates and increased respiration (Bao et al., 2019; Hapsari et al., 2025). They may also experience a change in their immunological function, which can be modeled as a change in hemocyte concentrations (Dittmer et al., 2011; Impellitteri et al., 2022).

Current studies have analyzed how metals such as copper, cadmium, and mercury have affected European Green Crabs (*Carcinus maenas*) in different temperatures (Truscott & White, 1990). Although a beneficial study, the effect of lead and copper has yet to be tested on smaller crabs, such as Purple Shore Crabs, in consistent, ambient temperatures. Thus, this study tests the physiological stress responses and immunological responses to lead and copper in ambient conditions. The copper and lead treatment groups were expected to experience changes in immune system function, slow behavioral reactions, and disrupted osmoregulation.

Methods

Experimental Setup

This study was conducted over the span of 3 weeks. Three treatments were used to determine the immunological and physiological stress responses in Purple Shore Crabs: control, lead, and copper. The weight of lead used was 29.18 g and the weight of copper used was 30.22 g. Similar weights of metals were used to keep treatments uniform. Each tank was kept at ambient conditions of 14°C and 35 ppt. 4 crabs were selected at random for each treatment (n=12). Feminized males were omitted from this study as the parasitic infection would have likely skewed the immunological response data and put the crabs at an increased risk of mortality (Dittmer et al., 2011). Individuals were tagged using bee tags, weighed, and had their carapace length and width measured. Baseline righting times were also recorded for each individual. For the duration of the experiment, crabs were starved. To decrease the likelihood of mortality, water changes were conducted every other day for the copper treatment. When copper ions dissipate, the water will change color, likely creating unnecessary stress for the crabs. Water changes were not done for the lead treatment because ions dissipating will not change the color of the water (Cardeilhac & Whitaker, 1988).

Parameters Tested

Righting Time

Righting time was determined by the amount of time it took each individual to flip upright after being turned on their backs. Righting time is a sufficient indicator of increased stress as more energy is being allocated to survival instead of movement (Maraschi et al., 2024; Taylor et al., 2014). Average righting time per treatment per week was graphed using RStudio (Figure 1) and analyzed using three separate one-way ANOVA's.

Total Hemocyte Count

To measure the total hemocyte count in each individual, hemolymph was extracted from each individual using a U-100 insulin syringe and inserted into the membrane of the third or fourth leg of each crab. About 1/8 of a milliliter was extracted from each individual. Hemolymph samples were kept in 15 µl of EDTA anti-coagulant until week 3 of the study. Samples were pipetted onto a Neubauer Hemocytometer slide under 40x magnification and stained using methylene blue. Hemocytes were identified based on pictures provided from Yildiz & Atar (2002), counted, and recorded. Total hemocytes found per treatment were graphed using RStudio (Figure 2) and analyzed using a one-way ANOVA.

Osmolality

Lastly, to measure osmolality, the hemolymph that was extracted for hemocyte counts was used. A small amount of each sample was put into an osmometer and documented. Average osmolality per treatment was graphed using RStudio (Figure 3) and analyzed using a one-way ANOVA. A second one-way ANOVA was performed to determine the statistical significance of the change in osmolality of the crabs in the lead treatment compared to the control treatment.

Results

Righting Time

A one-way ANOVA was used to test the effect of the different treatments on the individuals' righting times over the 3 weeks. This test concluded that heavy metals did not have a statistically significant effect on righting time per week (Control: p=0.0649, Lead: p=0.326, Copper: p=0.729).

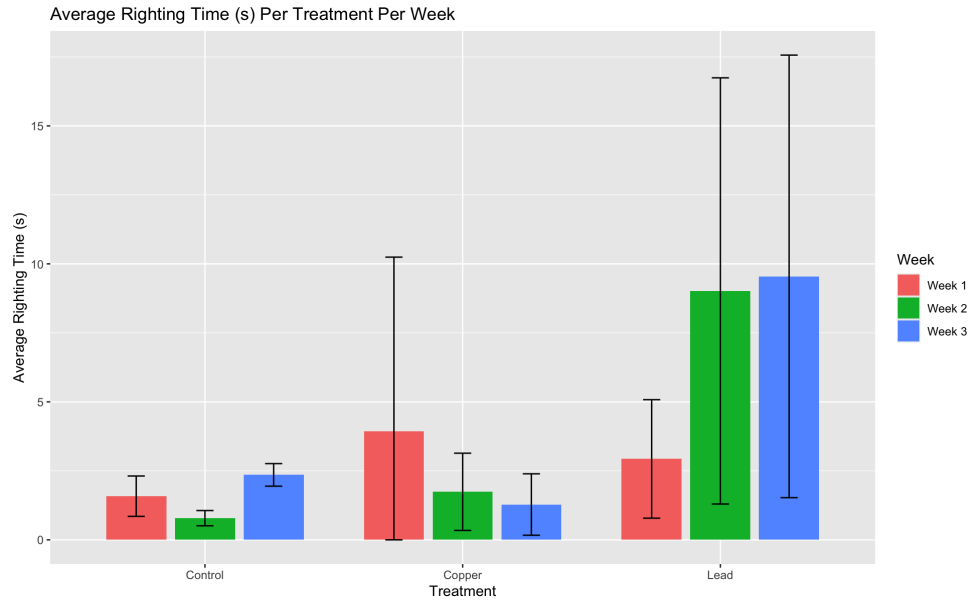


Figure 1. Average righting time (s) per treatment (control, copper, lead) over three weeks. Bar graph with standard deviation error bars. Fastest average righting time found in control treatment. Variation in copper and lead treatments compared to control with lead having the slowest average righting time over the three weeks.

Total Hemocyte Count

A one-way ANOVA was utilized to determine the statistical significance of the effect of heavy metals on the total hemocyte count per treatment group. This test shows no statistically significant effect, as $p=0.244$.

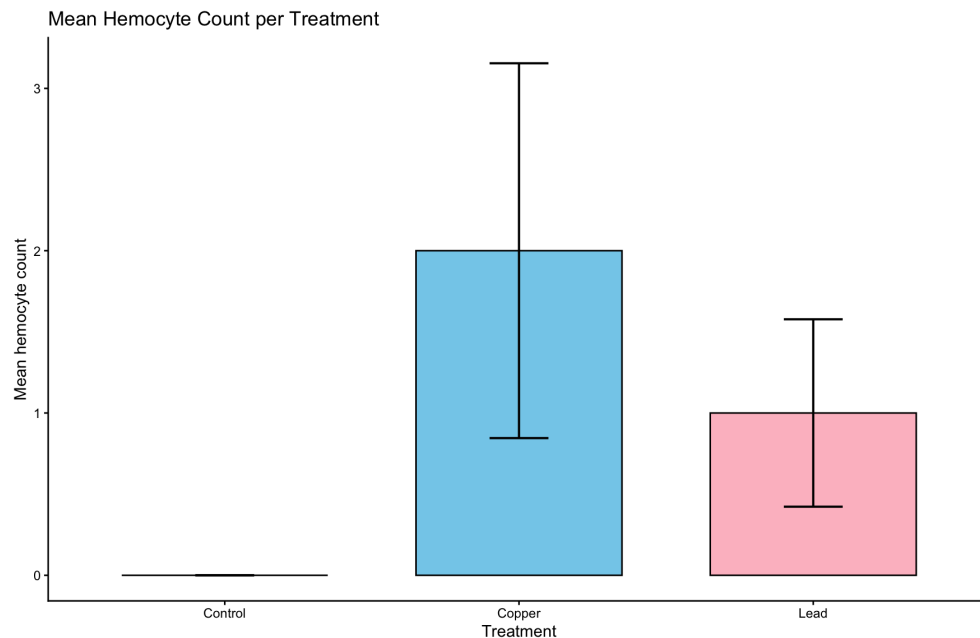


Figure 2. Mean hemocyte count per treatment. Bar graph with standard error bars of 95% confidence interval. Found no hemocytes in control treatment. Increased hemocyte count in both copper and lead treatments with copper having the highest mean count.

Osmolality

A one-way ANOVA was used to determine the effect of heavy metals on the mean osmolality of each treatment group. This test shows no statistically significant results, as $p=0.0633$. A second one-way ANOVA was performed to determine the statistical significance of the difference in osmolality from the lead treatment compared to the control group. This test shows a statistically significant result, with $p=0.0065$.

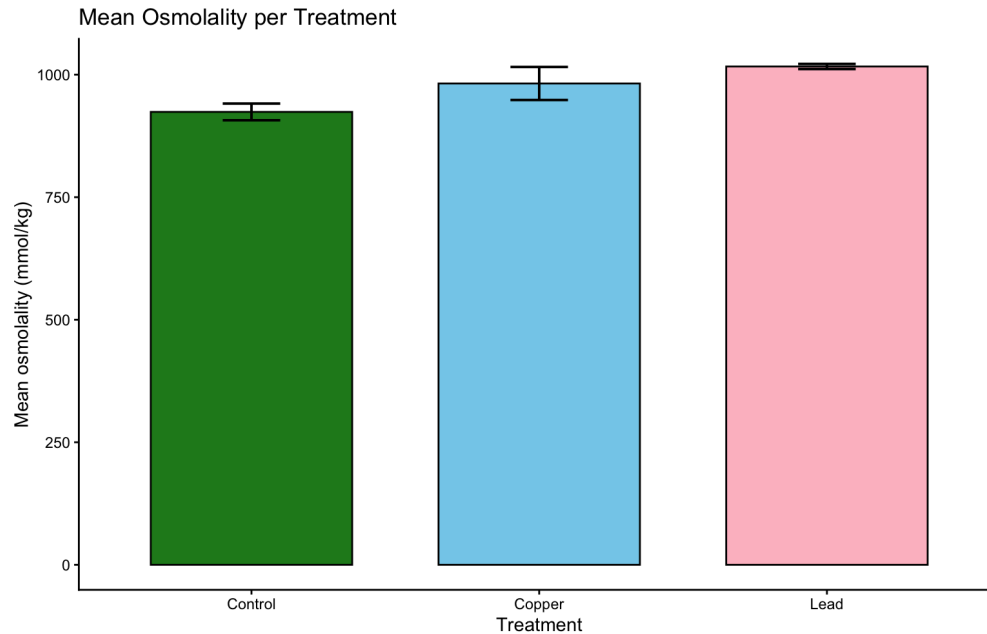


Figure 3. Mean osmolality (mmol/kg) per treatment. Bar graph with standard error bars of 95% confidence interval. Control treatment shows the lowest mean osmolality. Copper and lead treatments show increased osmolality compared to control treatment with lead having the highest mean values.

It is important to note that in the last week of trials, one crab from each treatment group died. These crabs were not included in the results of osmolality or righting times for week 3.

Discussion

Righting Time

Righting time was determined by the amount of time (in seconds) it took an individual to flip upright after being flipped on its back. A one-way ANOVA was run to determine the effect of different heavy metals on the average righting times per week. Some key results to note are that although we found statistically insignificant results from the ANOVA, with each p-value being greater than 0.05, Figure 1 demonstrates an increase in average righting time for both copper and lead treatments compared to the control group. This increase in righting time likely indicates that exposure to heavy metals increases the physiological stress responses of Purple Shore Crabs.

Total Hemocyte Count

Although statistically insignificant results, hemocyte counts visualized in Figure 2 represent a possible significant change between treatments. The lack of significant results is likely due to the small sample size of $n=9$. With an increased sample size, a change in hemocyte counts would be easier to identify.

With the data collected, an increase in hemocytes may suggest an immunological stress response. When the crabs were exposed to different heavy metals, hemocyte counts increased to fight off the possible invader in their bodies (Dittmer et al., 2011; Impellitteri et al., 2022).

Osmolality

Osmolality represents the concentration of dissolved solutes in the hemolymph of these crabs (DiBartola, 2006). Based on the results, the mean osmolality of both copper and lead treatments increased compared to the control treatment. Although they did not increase statistically significantly, osmolality did still differ between treatments.

When comparing the mean osmolality of the lead treatment crabs compared to the control crabs, a statistically significant result was found. This result means that the crabs in the lead treatment are more physiologically stressed when compared to the control crabs.

Limitations

While only one statistically significant result was found, it's important to note the limitations of this study. During the third week of trials, one crab from each treatment died. These deaths decreased our replicates by three. With this decrease in replicates, statistically insignificant results became more likely.

The second limitation encountered included the majority of our hemolymph samples being centrifuged. We attempted to find hemocytes during week 2 with our centrifuged samples, but failed to do so. The centrifuge likely broke the cells into smaller pieces, making it difficult to identify their presence.

Conclusion

In summary, with more replicates, a significant increase in righting time, total hemocytes, and osmolality would likely have been observed. From these results, future studies should include more replicates, different weights of heavy metals to determine if the concentration of ions has any effect, and different species of crabs to determine the effect on the marine environment in full.

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